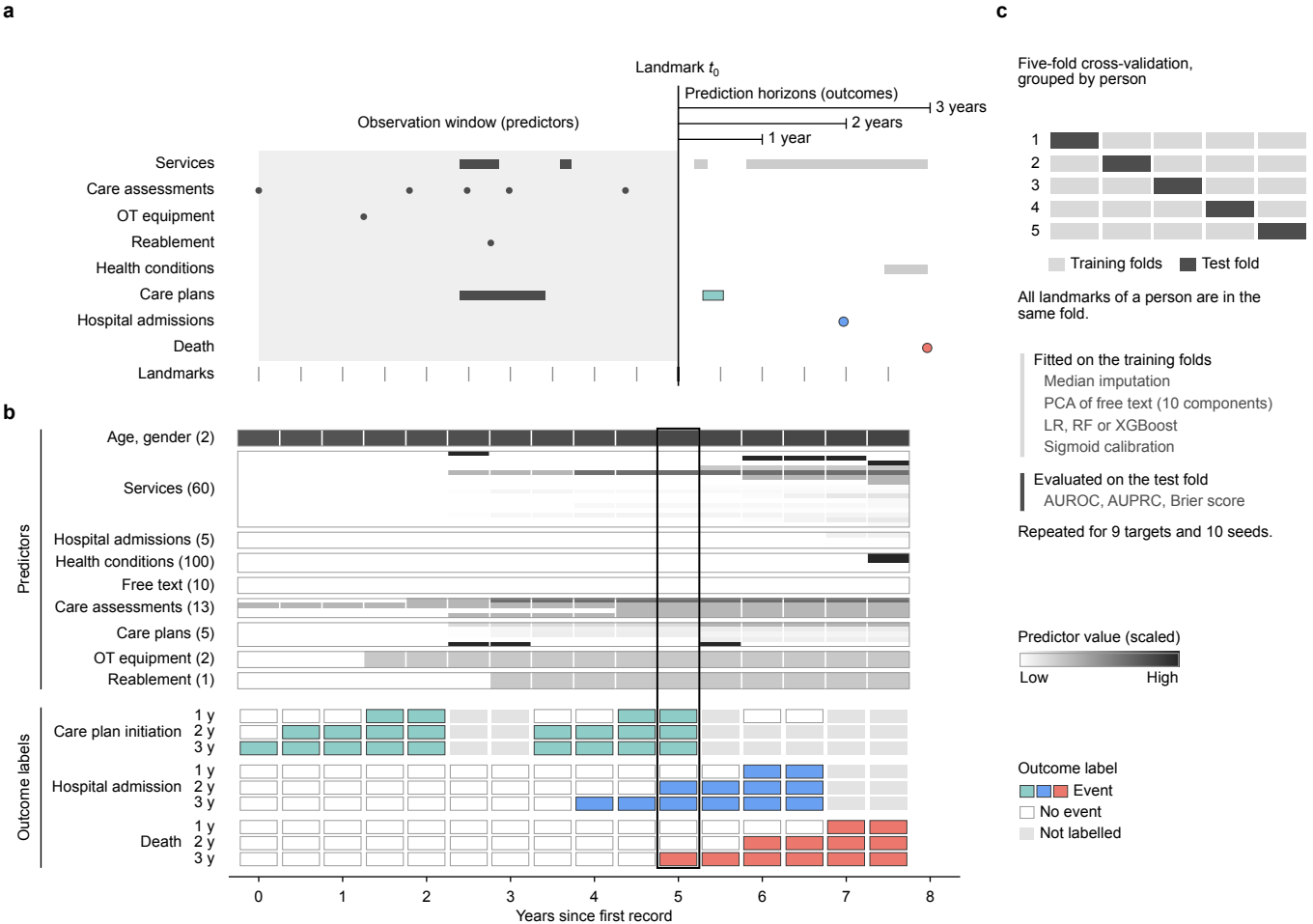
a
Landmark t_0
Prediction horizons (outcomes)
3 years
2 years
1 year
Observation window (predictors)
Services
Care assessments
OT equipment
Reablement
Health conditions
Care plans
Hospital admissions
Death
Landmarks
b
Predictors
Age, gender (2)
Services (60)
Hospital admissions (5)
Health conditions (100)
Free text (10)
Care assessments (13)
Care plans (5)
OT equipment (2)
Reablement (1)
Outcome labels
Care plan initiation
Hospital admission
Death
1 y
2 y
3 y
0
1
2
3
4
5
6
7
8
Years since first record
c
Five-fold cross-validation, grouped by person
1
2
3
4
5
Training folds
Test fold
All landmarks of a person are in the same fold.
Fitted on the training folds
Median imputation
PCA of free text (10 components)
LR, RF or XGBoost
Sigmoid calibration
Evaluated on the test fold
AUROC, AUPRC, Brier score
Repeated for 9 targets and 10 seeds.
Predictor value (scaled)
Low
High
Outcome label
Event
No event
Not labelled